Communicating the COVID-19 Crisis

A Comparative Analysis of Crisis Communication by Governments and Heads of State

Method Documentation

V2.0

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https://www.uibk.ac.at/en/political-science/research/polcom/projects/communicating-covid-19/
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1. Purpose of this Report

This report accompanies the study "Communicating the COVID-19 Crisis", a comparative analysis of crisis communication by governments and heads of state (https://www.uibk.ac.at/en/political-science/research/polcom/projects/communicating-covid-19/) and provides a detailed documentation of the data collection, the transcription and translation of press conference materials, and the preparation of the data for analysis.

2. Conditions of Use

Hayek, Lore; Dingler, Sarah C.; Senn, Martin; Schwaderer, Christian; Kraxberger, Andreas M.; Ragheb, Nada; Nordone, Fiona C., 2025, "Communicating the COVID-19 Crisis: A Comparative Analysis of Crisis Communication by Governments and Heads of State – Method Documentation (V2)", https://doi.org/10.11587/RWHCSF, AUSSDA, V2 is licensed under CC BY 4.0.

3. Study Description

Title: Communicating the Covid-19 Crisis

Researchers: Dr. Lore Hayek; Dr. Martin Senn; Dr. Sarah C. Dingler; Christian Schwaderer; Nada Ragheb; Andreas M. Kraxberger; Fiona C. Nordone

Collaborators: IUI Innsbruck University Innovations

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Project number: P 34225

File names:

- commcovid_data_segments, Hauptdatensatz
- commcovid_press_conference
- commcovid_speakers_biodata

4. Version Updates

- Addition of 46 new observations: press conferences by the White House Coronavirus Task Force of the United States Department of State, from January 31, 2020, to April 27, 2020
- Adaptation of country codes according to ISO-3166 format (SK to KR; UK to GB)
- Minor corrections to the "url" variable in both the commcovid_press_conferences and commcovid_data_segments datasets
- Minor corrections to the "pc_date" variable in both the commcovid_press_conferences and commcovid_data_segments datasets
- Minor corrections to the "file_name" variable in the commcovid_press_conferences dataset
- Minor corrections to the "doc-id" variable in the commcovid_data_segments dataset

5. Study Design

5.1. Sampling and Analysis Time Frame:

The study seeks to examine the crisis communication strategies employed by different governments in their respective press conferences and televised addresses, namely the governments of 18 OECD countries (Austria, Belgium, Canada, Chile, Czech Republic, France, Germany, Hungary, Iceland, Israel, Italy, New Zealand, South Korea, Spain, Sweden, Turkey, the UK and the US), as well as 3 US States (New York, Vermont, and Iowa).

The study looks at televised press conferences and public addresses that took place during the first phase of the Covid-19 crisis, starting from the first public address held on Covid-19 up to the first public address, in which a government official announced a relaxation of the restrictive measures that had been put in place to contain the virus. Overall, the data covers a timespan from January 20, 2020 to July 19, 2020.

The following table presents an overview of the first assessment of the collected data.

Country/ State Code	Country/ state	no. of press conferences	no. of televised addresses and appearances	Total number of public addresses	no. of speakers*	**first confirmed case**	first address	first announcement of relaxation of measures	total length of video material in h
AT	Austria	66	2	68	53	**26.02.2020**	26.02.2020	03.05.2020	43,65
BE	Belgium	39	1	40	13	**04.02.2020**	13.03.2020	23.04.2020	11,05
CA	Canada	44	0	44	1	**26.01.2020**	21.02.2020	28.04.2020	6,99
CL	Chile	28	6	34	4	**24.02.2020**	13.03.2020	19.07.2020	5,84
CZ	Czech Republic	9	0	9	10	**01.03.2020**	15.03.2020	20.04.2020	12,71
DE	Germany	14	12	26	10	**28.01.2020**	11.03.2020	30.04.2020	7,89
ES	Spain	78	5	83	38	**31.01.2020**	12.03.2020	28.04.2020	30,84
FR	France	13	5	18	11	**24.01.2020**	02.03.2020	12.07.2020	10,73
HU	Hungary	60	2	62	23	**02.03.2020**	05.03.2020	29.04.2020	18,42
IL	Israel	25	1	26	17	**21.02.2020**	02.02.2020	04.05.2020	9,18
IS	Iceland	46	1	47	52	**28.02.2020**	26.02.2020	15.04.2020	26,54
IT	Italy	25	8	33	13	**21.02.2020**	30.01.2020	16.05.2020	10,44
NZ	New Zealand	73	4	77	20	**28.02.2020**	21.01.2020	27.04.2020	34,69
SE	Sweden	75	2	77	59	**31.01.2020**	31.01.2020	13.05.2020	25,01
KR	South Korea	336	0	336	47	**20.01.2020**	20.01.2020	21.04.2020	160,10
TR	Turkey	13	10	23	2	**11.03.2020**	18.03.2020	06.05.2020	6,41
GB	United Kingdom	52	5	57	27	**31.01.2020**	03.03.2020	13.05.2020	42,39
IA	Iowa	40	5	45	29	**03.03.2020**	08.03.2020	18.05.2020	22,57
NY	New York	49	0	49	14	**01.03.2020**	26.02.2020	15.04.2020	19,25
VT	Vermont	13	0	13	18	**07.03.2020**	05.03.2020	24.04.2020	13,29
US	United States of America	46	0	46	74	**20.01.2020**	31.01.2020	27.04.2020	0,00
Total	-	1144	69	1213	535	-	-	-	517,99

*based on collected bio data, subject to corrections

5.2. Primary Data Collection

5.2.1. Raw Data

For each country all available videos and speech transcripts were collected from official government sources, wherever possible, and news agencies, wherever no official government source was found. Both the addresses by heads of government and heads of state were taken into account, as well as press conferences held by Health Ministry officials and/or expert panels respectively. The following table presents a summary of the sources used for each country respectively.

Country/ State Code	Country/ state	Source(s)
AT	Austria	(1) Official Youtube Channel of the Federal Chancellorship: https://www.youtube.com/channel/UC-R71rozmdFPQCR9Mi1Zy5g (2) Official Youtube Channel of Alexander van Bellen: https://www.youtube.com/channel/UCguMeO2ZTxHB0ytM0b_1NRw
BE	Belgium	(1) Official Youtube Channel of the Federal Health Authority in Dutch: https://www.youtube.com/user/gezondheidbelgie/featured (2) Official Youtube Channel of the Federal Health Authority in French: https://www.youtube.com/user/santebelgique/featured (3) Youtube Channel of a Royal Correspondent, Rick Evers: https://www.youtube.com/c/RickEvers/featured (4) Official Youtube Channel of the Residential Palace: https://www.youtube.com/channel/UCbtgsBjTqWypAO8210-FeyQ/featured (5) Youtube Channel "IPC Videohouse": https://www.youtube.com/channel/UCEs5d-nmvowbxGe54_Gobqw/featured
CA	Canada	(1) Video Archive - Official Website of Prime Minister Justin Trudeau: https://pm.gc.ca/en/videos (2) Official Youtube Channel of Prime Minister Justin Trudeau: https://www.youtube.com/channel/UCyvEONnqE2Krm9Zi0LVvGmA
CL	Chile	Video Archive - Official Website of the Chile Presidency: https://prensa.presidencia.cl/videos.aspx
CZ	Czech Republic	Official Youtube Video of the Czech Press Office: https://www.youtube.com/channel/UC1x5byV217xZZFDBQOJQSOg

DE	Germany	(1) Coronavirus-related Update Page - Official Website of the German Federal Government: https://www.bundesregierung.de/breg-de/themen/coronavirus/ (2) Video Archive - Official Website of the German Federal Government: https://www.bundesregierung.de/breg-de/service/archiv/archiv-mediathek/ (3) Official Youtube Channel of the German Federal Government: https://www.youtube.com/channel/UCcnFmkINMo4Jmkx5mvlCh6g (4) Speech Archive - Official Website of the German Presidency: https://www.bundespraesident.de/DE/Bundespraesident/Reden-und-Interviews/Reden/reden-node.html;jsessionid=EF899B0E1E5D55A260B635E15AED92D2.2_cid361
ES	Spain	(1) Video Archive - Official Website of the Spanish Government: https://www.lamoncloa.gob.es/multimedia/videos/covid19/Paginas/index.aspx (2) Official Youtube Channel of the Spanish Government: https://www.youtube.com/channel/UCzJtDQ9b_Uq1tA5W_tMiMUQ (3) Speech Archive - Official Website of the Spanish Royal Residence: https://www.casareal.es/ES/Actividades/Paginas/actividades_discursos.aspx (4) Official Youtube Channel of the Spanish Royal Family: https://www.youtube.com/user/casarealtv/featured
FR	France	(1) Official Youtube Channel of the French Presidency: https://www.youtube.com/channel/UCPaeEhnVIdn4T01gUphIltw (2) Official Youtube Channel of Emmanuel Macron: https://www.youtube.com/channel/UCFqGa9uitcB-fWyNZK2xImw (3) Updates Page - Official Website of the French Presidential Residence: https://www.elysee.fr/toutes-les-actualites (4) Speech Archive Search - Official Website of the French Government: https://www.gouvernement.fr/discours-et-rapports
HU	Hungary	(1) Official Youtube Channel of the Hungarian Government: https://www.youtube.com/channel/UCHBJUOq1BvR7ptPJ_MAMvbg (2) Speech Archive - Official Website of the Prime Minister's Office: https://www.miniszterelnok.hu/#beszedek_container (3) Speech Archive - Official Website run by the Prime Minister's Office: https://abouthungary.hu/speeches-and-remarks (4) Speech Archive - Official Website of the Hungarian Presidency: https://www.keh.hu/beszedek/ (4) Various News Articles from "Index" News Agency: https://index.hu/belfold/# (5) Official Facebook Page of Viktor Orban: https://www.facebook.com/orbanviktor

IL	Israel	(1) News page - Official Website of the Israeli Government: https://www.gov.il/en/departments/news/ (2) Official Youtube Channel of the Israeli Prime Minister: https://www.youtube.com/channel/UC4XJnRPZjXhgvVMhXKNSJvQ
IS	Iceland	(1) Various News Articles from "visir" News Agency: https://www.visir.is/ (2) Various News Articles from "ruv" News Agency: https://www.ruv.is/ (3) Speech Archive - Official Website of the Icelandic Presidency: https://www.forseti.is/
IT	Italy	(1) Official Youtube Channel of the Italian Government: https://www.youtube.com/c/palazzochigi/featured (2) Official Youtube Channel of the Italian Presidency: https://www.youtube.com/channel/UCWVTngiI0h4XpCB_nKtdY7g (3) Speech Archive - Official Website of the Italian Presidency: https://www.quirinale.it/ricerca/discorsi
NZ	New Zealand	(1) Official Youtube Channel of New Zealand's Ministry of Health: https://www.youtube.com/channel/UCPuGpQo9IX49SGn2iYCoqOQ (2) Official Youtube Channel of RNZ News: https://www.youtube.com/c/RNZVideo/featured (3) Speech Archive - Official Website of the New Zealand Government: https://www.beehive.govt.nz/speeches (4) Official Youtube Channel of the Governor General of New Zealand: https://www.youtube.com/channel/UCXkEAWxqC3FfHs8r3f7NISQ (5) Official Youtube Channel of Newhub: https://www.youtube.com/channel/UCWq564ONoREJprMNhewS6Sw
SE	Sweden	(1) Official Youtube Channel of the Swedish Government: https://www.youtube.com/channel/UCTCf9DNzLC78u2o_4Iu2frw (2) Official Youtube Channel of the Swedish Royal Residence: https://www.youtube.com/channel/UCZPxZX-eRWxvEt2n65UstOA
KR	South Korea	(1) Official Youtube Channel of the South Korean Government: https://www.youtube.com/user/chKTV520/featured (2) Official Website of the Korean Presidency: https://www.president.go.kr/
TR	Turkey	(1) Official Youtube Channel of Recep Tayyip Erdogan: https://www.youtube.com/channel/UCiHR69jCQX8uoH-B1tX6lzg (2) Speech Archive - Official Website of the Turkish Presidency: https://www.tccb.gov.tr/receptayyiperdogan/konusmalar/
GB	United Kingdom	(1) Official Youtube Channel of the UK Government: https://www.youtube.com/channel/UC8o7mIMg3mmO9-dx3e2iFgw (2) Official Youtube Channel of the British Royal Family:

		https://www.youtube.com/channel/UCTkC3Jt91QkqNAE4FGWkEIQ (3) Official Website of the British Royal Family: https://www.royal.uk/
US	United States of America	(1) Official Website of the Trump White House Archive: https://trumpwhitehouse.archives.gov/ (2) The American Presidency Project: https://www.presidency.ucsb.edu/documents
IA	Iowa	Official Facebook Page of Iowa Governor Kim Reynolds: https://www.facebook.com/watch/IAGovernor/
NY	New York	Official Youtube Channel of Former Governor Andrew Cuomo: https://www.youtube.com/channel/UCLHU6ECVSZbzgcpd113Jk4Q
VT	Vermont	Official Youtube Channel of ORCA Media: https://www.youtube.com/c/ORCAMedia/featured

For a more detailed description of the variables see Hayek, Lore; Dingler, Sarah C.; Senn, Martin; Schwaderer, Christian; Kraxberger, Andreas M.; Ragheb, Nada; Nordone, Fiona C., 2025, "Communicating the COVID-19 Crisis: A Comparative Analysis of Crisis Communication by Governments and Heads of State – Codebook", https://doi.org/10.11587/RWHCSF, AUSSDA, V2

5.2.2. Meta Data

For each file, the corresponding meta data was recorded, which included details about the file itself as well as information pertaining to the setting and characteristics of the event.

- unique document ID: Videos were downloaded and archived, each receiving a unique ID that consists of the following information, illustrated below using an example from the metadata file.: FR_20200413_01

  XX...Two Letter ISO Code for the Country/US State the video pertains to.
  YYYYMMDD...Date of Press Conference, where the date was unclear, the date of upload was used as an indicator instead.
  01...Running number, to differentiate multiple videos uploaded in one day

  The example above is therefore clearly identifiable as the transcript of a press conference from France broadcast on the 13th of April 2020.

Information regarding each event includes:

- the *type of event,*
- the *date* of the event,
- the *time* of the event (where available),

- a *Q&A indicator* to track whether an event included a Q&A session,
- the *languages* used at the event,
- the *state code,*
- the *speakers* who took part in the event,
- the *function* of each speaker with regards to the pandemic or government office,
- the *gender* of each speaker,
- the total *number of speakers* at the event,
- an indicator on whether any *safety measures* were taken during the event,
- a *description of the safety measures* taken (where applicable),
- as well as the number of *views on Youtube* (at the time of download).

Also included are details about the file itself, such as

- the *total length of the file*,
- the *type of file*,
- the *download url*,
- the *download date,*
- *timestamps* for each change in language,
- and *Q&A timestamps* marking when the Q&A session begins,
- as well as additional *notes* (mostly of technical nature)

4.2.3. Transcription and Translation

As mentioned above, the transcription of videos and translation of all gathered data into English was conducted by Innsbruck University Innovations (IUI). On the basis of the raw and metadata described above, a finalized data set was created that is made up of textual data; one transcript for each event. To achieve this, the following steps were taken:

- Split
  The Q&A timestamps were used to automatically split up all the press conference videos, where the Q&A part was dismissed. For this the free software ffmpeg was used (https://ffmpeg.org/). So we end up with one or more files for each press conference, each containing only one language.

- Convert
  Then ffmpeg was also used to convert all these videos to flac (https://xiph.org/flac/), which is a lossless audio format. Using flac not only speeds up the speech recognition, it also results in a better recognition quality.

- Upload to gcs
  The google cloud Speech2Text service needs audio files longer than one minute to be present on the

google cloud storage service, so all the flac files were uploaded to it, using the gsutil tool (https://cloud.google.com/storage/docs/gsutil?hl=de).

- Speech2text Recognition
  After that the gathered language metadata was used to start the Speech2Text recognition for each flac file. The Google Cloud Service API was accessed using Java (https://java.com/).

- Created 3 Output Formats
  The Speech2Text recognition results were written into 3 different output files, a plain text file, a plain text file with time marks every 30 seconds, to be able to quickly check against the video, and a CSV file, containing the time marks for every recognized word. In the case of multilingual press conferences, the different language parts had to be merged together into one output file.

- Quality Measures
  At this point a cross reference between video files and result files was performed and we took samples to determine the quality of the recognition, which led to the reprocessing of several videos with different language metadata to improve the quality. The runtime of the conversion and the recognition together was about one third of the video's runtime, leading to several weeks processing time which made extended process management and handling necessary.

- Convert to Word
  The plain text files, with the time marks, were converted to word documents (.docx), because SmartCat (https://www.smartcat.com/), the used translation platform, has better support for them.

- Upload to SmartCat
  Then those word documents were uploaded to SmartCat, which was used to distribute them to all the invited language-specific freelancers.

- Transcription Proofreading
  First, the automatically created transcripts were given to language-specific transcribers, who corrected them according to the original video material.

- Machine Translation
  The corrected transcripts were then machine translated on SmartCat, which offers the choice between several providers. When possible the texts were translated with DeepL; for all the languages not available on DeepL (Hebrew, Icelandic, Norwegian, Korean, Turkish) Google Neural Machine Translation was used.

- Post-editing
  The machine-translated texts were then again given to freelancers on SmartCat who corrected them. Whenever possible, these post-editors were native speakers of English who are currently living in or have at least lived for some time in a country where the target language is spoken.

- Final Delivery
  The final product were the corrected texts and these were delivered together with the corrected transcripts in the respective original language in Word format.

The transcripts delivered by IUI were imported and matched with the corresponding metadata to form one final data.frame for analysis.